

Azure Cloud ClickOps – Basic to Expert Level Fully Loaded Documentation

Learning Path Structure

Level 1: Beginner (Azure Fundamentals)

- Azure Portal navigation
- Subscriptions
- Resource Groups
- Basic VNets
- VM deployment
- Storage basics
- User creation in Entra ID
- Cost basics
- Monitoring basics

Level 2: Intermediate (Administration & Operations)

- Advanced networking
- NSGs
- Load Balancers
- App Services
- Backup
- RBAC
- Policy management
- Azure Firewall
- Hybrid connectivity
- Operational governance

Level 3: Advanced (Architecture & Security)

- Landing zones
- Multi-region architecture
- Hub-spoke networking
- Zero Trust
- Defender for Cloud
- Sentinel SIEM
- PIM
- Conditional Access
- DR strategy
- FinOps optimization

Level 4: Expert (Enterprise Cloud Leadership)

- Multi-cloud governance
- IaC transformation
- Terraform integration
- DevSecOps

- Compliance frameworks
 - Enterprise security operations
 - Cost governance at scale
 - Cloud migration frameworks
 - Executive reporting
 - Strategic architecture planning
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Overview

This guide is designed for infrastructure professionals, cloud administrators, and architects who manage Microsoft Azure environments using **ClickOps** (manual configuration through the Azure Portal GUI).

ClickOps = Managing Azure resources through the web portal instead of Infrastructure as Code (Terraform, ARM, Bicep, etc.).

1. Azure Portal Fundamentals

Core Areas:

- Azure Dashboard
- Resource Groups
- Virtual Networks (VNet)
- Virtual Machines (VMs)
- Storage Accounts
- Network Security Groups (NSGs)
- Azure Firewall
- Load Balancers
- App Services
- Azure Active Directory (Microsoft Entra ID)
- Cost Management
- Monitoring
- Backup & Disaster Recovery

Common Administrative Workflow:

1. Login to Azure Portal
 2. Select Subscription
 3. Create Resource Group
 4. Deploy Resources
 5. Configure Security
 6. Enable Monitoring
 7. Set RBAC
 8. Backup Resources
 9. Optimize Costs
 10. Document Changes
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2. Azure Landing Zone Foundation (Enterprise Cloud Base)

What is a Landing Zone?

An Azure Landing Zone is a secure, governed, scalable cloud foundation designed before deploying workloads.

Core Purpose:

- Governance
- Security
- Identity
- Networking
- Compliance
- Cost control
- Operational consistency

Landing Zone Core Components:

Management Groups:

- Root
- Platform
- Production
- Non-Production
- Sandbox

Subscription Strategy:

- Identity Subscription
- Connectivity Subscription
- Management Subscription
- Production Subscription
- Development Subscription

Identity:

- Microsoft Entra ID
- MFA
- RBAC
- PIM
- Conditional Access

Networking:

- Hub-Spoke topology
- Azure Firewall
- Bastion
- VPN Gateway

- ExpressRoute
- DNS

Security:

- Defender for Cloud
- Sentinel
- Key Vault
- NSGs
- DDoS Protection
- Private Endpoints

Governance:

- Azure Policy
- Tagging
- Naming conventions
- Budgets
- Locks
- Compliance controls

Monitoring:

- Azure Monitor
- Log Analytics
- Security Center
- Cost Management

Enterprise Landing Zone Workflow:

1. Create Management Groups
2. Define Subscription hierarchy
3. Configure identity baseline
4. Build networking architecture
5. Deploy security controls
6. Apply governance policies
7. Enable monitoring
8. Establish backup & DR
9. Configure budgets
10. Document architecture

Professional Objective:

Landing Zone ensures enterprise cloud deployments are standardized, secure, scalable, and compliant.

3. Subscription & Governance Setup

Steps:

Create Management Group:

- Search → Management Groups
- Create hierarchy for:
 - Production
 - Staging
 - Development

Assign Policies:

- Allowed Regions
- Tagging Requirements
- Resource Type Restrictions
- Security Baselines

Role-Based Access Control (RBAC):

- Owner
- Contributor
- Reader
- Network Contributor
- Security Admin

Best Practices:

- Apply least privilege
 - Enable MFA
 - Use PIM (Privileged Identity Management)
 - Enable Activity Logs
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4. Resource Group Deployment

Steps:

1. Search Resource Groups
2. Click Create
3. Select Subscription
4. Name Convention:
5. rg-prod-network
6. rg-dev-app
7. Choose Region

8. Apply Tags:

9. Environment

10. Owner

11. CostCenter

5. Virtual Network (VNet) Deployment

Steps:

Create VNet:

- Address Space: 10.0.0.0/16
- Subnets:
 - Web: 10.0.1.0/24
 - App: 10.0.2.0/24
 - DB: 10.0.3.0/24
 - Bastion: 10.0.4.0/27

Configure:

- DNS Servers
- DDoS Protection
- NSG association
- Route Tables
- Peering

Security:

- Use private endpoints
 - Restrict public IPs
 - Enable Azure Firewall
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6. Virtual Machine Deployment

Windows/Linux VM ClickOps:

Steps:

1. Search Virtual Machines
2. Click Create
3. Select:
4. Resource Group

5. VM Name
6. Region
7. Availability Zone
8. Image Selection:
 9. Windows Server 2022
 10. Ubuntu 22.04
 11. RHEL
12. Size:
 13. B-series (testing)
 14. D-series (production)
15. Authentication:
 16. SSH Key
 17. Password
18. Networking:
 19. VNet/Subnet
 20. Public IP (optional)
 21. NSG
22. Monitoring:
 23. Boot Diagnostics
 24. Azure Monitor
25. Backup:
 26. Enable Recovery Vault

Post Deployment:

- Install extensions
 - Configure Defender
 - Apply Updates
 - Domain Join
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7. Storage Account Management

Types:

- Blob Storage
- File Shares
- Queues
- Tables

Deployment:

Options:

- Standard/Premium
- LRS/ZRS/GRS/GZRS
- Hot/Cool/Archive

Security:

- Disable public access
 - Private Endpoints
 - Firewall restrictions
 - SAS token controls
 - Lifecycle management
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8. Identity & Access Management (Entra ID)

Key Components:

- Users
- Groups
- Conditional Access
- Enterprise Apps
- MFA
- SSO
- Identity Protection

Tasks:

User Management:

- Create user
- Assign license
- Add groups
- Reset password

Security:

- Enable MFA
 - Configure Conditional Access:
 - Block legacy auth
 - Require compliant device
 - Geo restrictions
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9. Network Security

NSG Configuration:**Common Rules:**

- Allow 443
- Allow 22/3389 only via Bastion
- Deny Any Any inbound

Azure Firewall:

- DNAT
- Application rules
- Network rules
- Threat intelligence

Bastion:

- Secure RDP/SSH without public exposure

DDoS:

- Standard plan for critical VNets
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10. Backup & Disaster Recovery

Azure Backup:

- Recovery Services Vault
- VM Backup Policies
- File Share Backup
- SQL Backup

ASR (Azure Site Recovery):

- Region-to-region replication
- Failover testing

- DR drills

Best Practices:

- Daily backups
 - Immutable vaults
 - Cross-region restore
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11. Monitoring & Logging

Tools:

- Azure Monitor
- Log Analytics
- Application Insights
- Network Watcher
- Security Center (Defender for Cloud)

Key Metrics:

- CPU
- Memory
- Disk IOPS
- NSG flow logs
- Security alerts
- Cost anomalies

Alerts:

- CPU > 80%
 - Disk > 90%
 - Unauthorized login attempts
 - Budget thresholds
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12. Cost Optimization ClickOps

Methods:

- Azure Advisor recommendations
- Reserved Instances
- Spot VMs
- Auto-shutdown
- Rightsizing
- Storage tiering

Governance:

- Budgets
 - Cost Alerts
 - Resource locks
 - Tagging for chargeback
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13. Security Hardening Checklist

Mandatory:

- MFA
- Defender for Cloud
- NSGs
- Azure Firewall
- Private Link
- Disk Encryption
- Key Vault
- RBAC
- Patching
- Logging

Advanced:

- Sentinel SIEM
 - Zero Trust
 - JIT VM access
 - PIM
 - Secure Score improvement
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14. Real-World Production ClickOps Workflow

Example:

Deploy Secure Web App:

1. Create RG
2. Create VNet + Subnets
3. Deploy NSGs
4. Deploy App Service Plan
5. Configure Web App
6. Add Azure SQL
7. Private Endpoint
8. Key Vault secrets
9. WAF via Application Gateway
10. Enable Monitoring

11. Configure Backup
 12. Assign RBAC
 13. Budget controls
 14. Security review
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15. ClickOps Risks

Challenges:

- Manual drift
- Human error
- Inconsistent deployments
- Poor scalability
- Weak auditability

Mitigation:

- Standard Operating Procedures (SOPs)
 - Naming conventions
 - Documentation
 - Change management
 - Transition to IaC over time
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16. Azure ClickOps to DevOps Transition

Progression:

Stage 1:

Portal administration

Stage 2:

CLI / PowerShell

Stage 3:

ARM/Bicep

Stage 4:

Terraform

Stage 5:

CI/CD Automation

17. Interview-Oriented Practical Skills

Expected Knowledge:

- VNet design
- NSGs
- RBAC
- Backup
- Monitoring
- Cost control
- Security Center
- Entra ID
- Load balancing
- Firewall
- DR

Scenario Questions:

- Secure production landing zone
- Hybrid connectivity
- Identity governance
- Cost optimization
- Incident response

18. Documentation Templates for Enterprise Use

Maintain:

- Resource inventory
- IP schema
- Security baseline
- DR plans
- SOPs
- Cost reports
- Compliance reports
- Access reviews

Final Professional Recommendation

For enterprise market standards:

Master:

- Azure Portal ClickOps

- Azure CLI
- PowerShell
- Terraform
- Security Architecture
- Governance
- FinOps
- DevOps automation

Market Reality:

ClickOps is useful for:

- Learning
- Troubleshooting
- Small deployments
- Emergency changes

Enterprise Standard:

- IaC + Governance + Security Automation
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Career Positioning Tip

If presenting yourself professionally:

“Experienced in Azure cloud administration through both ClickOps and Infrastructure as Code methodologies, including secure landing zone design, governance, networking, RBAC, backup, cost optimization, and enterprise-scale operational management.”

Conclusion

This document provides a practical end-to-end Azure ClickOps operational framework suitable for:

- Cloud Administrators
- Infrastructure Architects
- Security Engineers
- DevOps Transition Professionals

It forms the foundational layer before advanced automation and cloud architecture maturity.